



SARVAJANIK
UNIVERSITY

INCLUSIVE | INTEGRATED | INNOVATIVE

Sarvajani Education Society
Sarvajani College of Engineering & Technology, Surat.
Department of Computer Applications
Academic Year: 2025 – 26



A Project On “InvestSphere”

Presented By
Anant Soni (ET24MTCA064)

M.C.A. II Sem. 4

Under the Guidance of
Prof. Zankhana Vaishnav

AGENDA

- Company Profile
- System Description
- Project Overview
- Database Design
- File System
- Use Case Diagram
- Class Diagram
- Interaction Diagram
- Activity Diagram
- Data Dictionary
- Screen Shots
- References

Company Profile:

CODING HOUSE – Company Overview

CODING HOUSE is a technology-driven software development company focused on building reliable, scalable, and user-centric digital solutions. We help startups, small businesses, and growing enterprises turn ideas into high-quality products using modern technologies and proven development practices.

What We Do:

- Mobile App Development (Flutter, Android, iOS)
- Web Development (React, Node.js, PHP, WordPress)
- UI/UX Design (Figma-based, conversion-focused designs)
- Custom Software Solutions
- API Development & Integrations

Our Approach:

At CODING HOUSE, we follow a transparent and collaborative development process:

- Clear understanding of client requirements
- Structured planning and architecture
- Agile development with regular updates

System Description

Development Framework:

- Frontend developed using React.js
- Backend built with Node.js and Express.js
- RESTful APIs for seamless client-server communication

UI / Frontend Design:

- Component-based architecture using React.js
- Styled using Tailwind CSS for modern, responsive UI
- Mobile-friendly and optimized user experience

Programming Language:

- JavaScript (ES6+) used across frontend and backend
- Ensures consistency and faster development

Database:

- MongoDB Atlas (Cloud Database) used for data storage
- Stores:
 - User data
 - Investments Watchlists
- Scalable and accessible from anywhere

Media Storage:

- Cloudinary used for storing images and media files
- Fast CDN delivery and optimized asset handling

Authentication & Security:

- OTP-based email authentication system
- Secure login and user verification
- JWT (JSON Web Token) for session management

Project Overview:

- InvestSphere is a MERN-based web application for managing and tracking investments efficiently.
- It provides a centralized platform to monitor **Stocks, SIPs, IPOs, Market News and Watchlists** with real-time insights.
- Built on a **modular and scalable architecture**, ensuring high performance and maintainability.
- Includes secure authentication, data visualization, and interactive dashboards.
- Designed using scalable MERN architecture with cloud integration (MongoDB Atlas & Cloudinary)

Core System Features

- OTP-based Login & Registration
- Dashboard for SIP, Stocks & IPO tracking
- Watchlist with real-time market data
- Reports & Analytics for performance tracking
- Profile management with Cloudinary (image upload)
- Data stored securely in MongoDB Atlas
- API-based communication (Frontend ↔ Backend)

User Functional Capabilities

- Manage investments including SIP, Stocks, and IPOs
- Add and track assets in a personalized watchlist
- View real-time market data and insights
- Access reports and analytics for better decision-making
- Update profile details and upload images using Cloudinary integration
- Secure authentication using OTP & JWT

Database Design:

Database Name:

investsphereDB

Collection Name:

1. Register (Users Table)
2. Reports
3. Watchlists

Collection: Register (Users Table)

Field Name	Data Type	Description
_id	ObjectId	Unique user ID
name	String	User full name
email	String	User email address
password	String	Encrypted password
phone	String	Contact number
country	String	User country
State	String	User state
Dob	Date	Date of birth
Pan	String	PAN card number
Verified	Boolean	Email verification status
profileImage	String	Cloudinary image URL
profileImageId	String	Cloudinary public ID
createdAt	Date	Account creation date
updatedAt	Date	Last update date

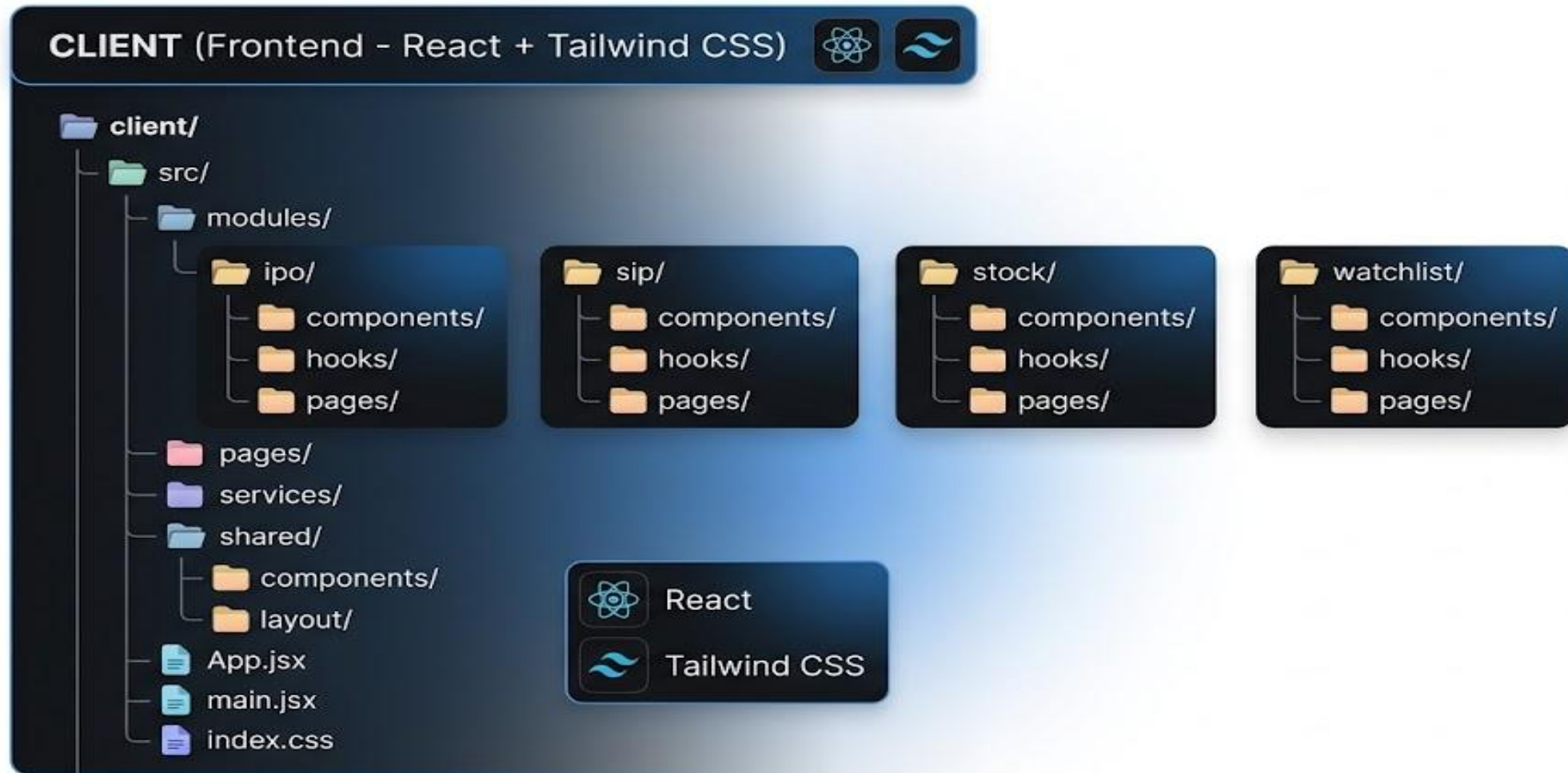
Collection: Reports

Field Name	Data Type	Description
_id	ObjectId	Unique report ID
title	String	Issue title
description	String	Issue description
userEmail	String	User email
username	String	User name
createdAt	Date	Report created date
updatedAt	Date	Last updated date

Collection: Watchlists

Field Name	Data Type	Description
_id	ObjectId	Unique ID
email	String	User email
itemCode	String	Asset code (IPO/Stock)
itemName	String	Asset name
type	String	Type (IPO / Stock / SIP)
createdAt	Date	Added date
updatedAt	Date	Last updated date

File System: (Client)



File System: (Server)

SERVER (Backend - Node.js + Express)



server/

config/

JS db.js

JS cloudinary.js

controllers/

JS authController.js

JS userController.js

routes/

JS authRoutes.js

stockRoutes.js

sipRoutes.js

watchlistController.js

utils/

JS server.js

.env

models/

JS userModel.js

JS watchlistModel.js

JS reportModel.js

middleware/

JS securityMiddleware.js

JS multer.js

JS stockRoutes.js

JS watchlistRoutes.js

services/

JS emailService.js

JS captchaService.js

JS websocketService.js

server.js

JS media.app.js

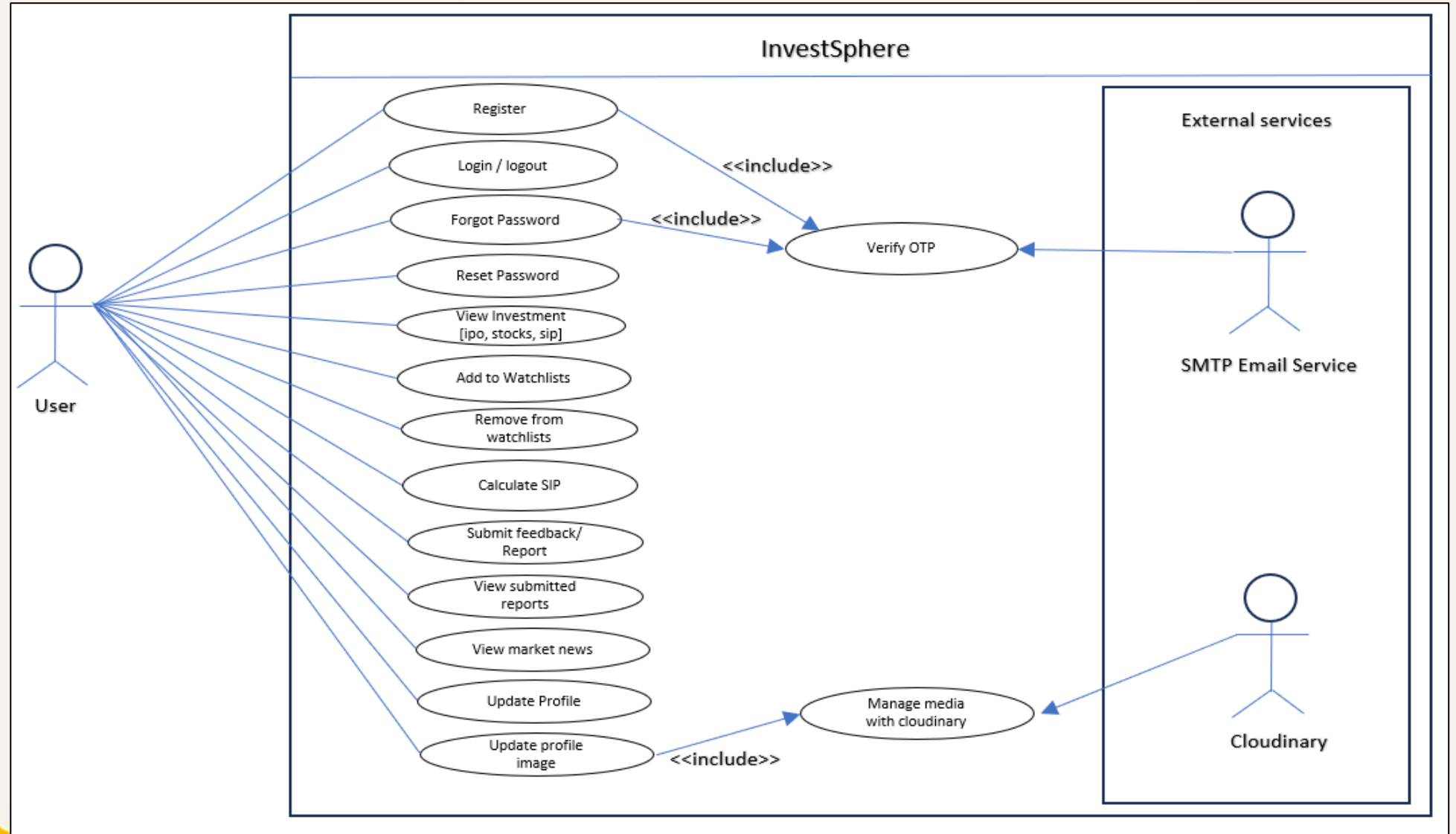


MongoDB Atlas

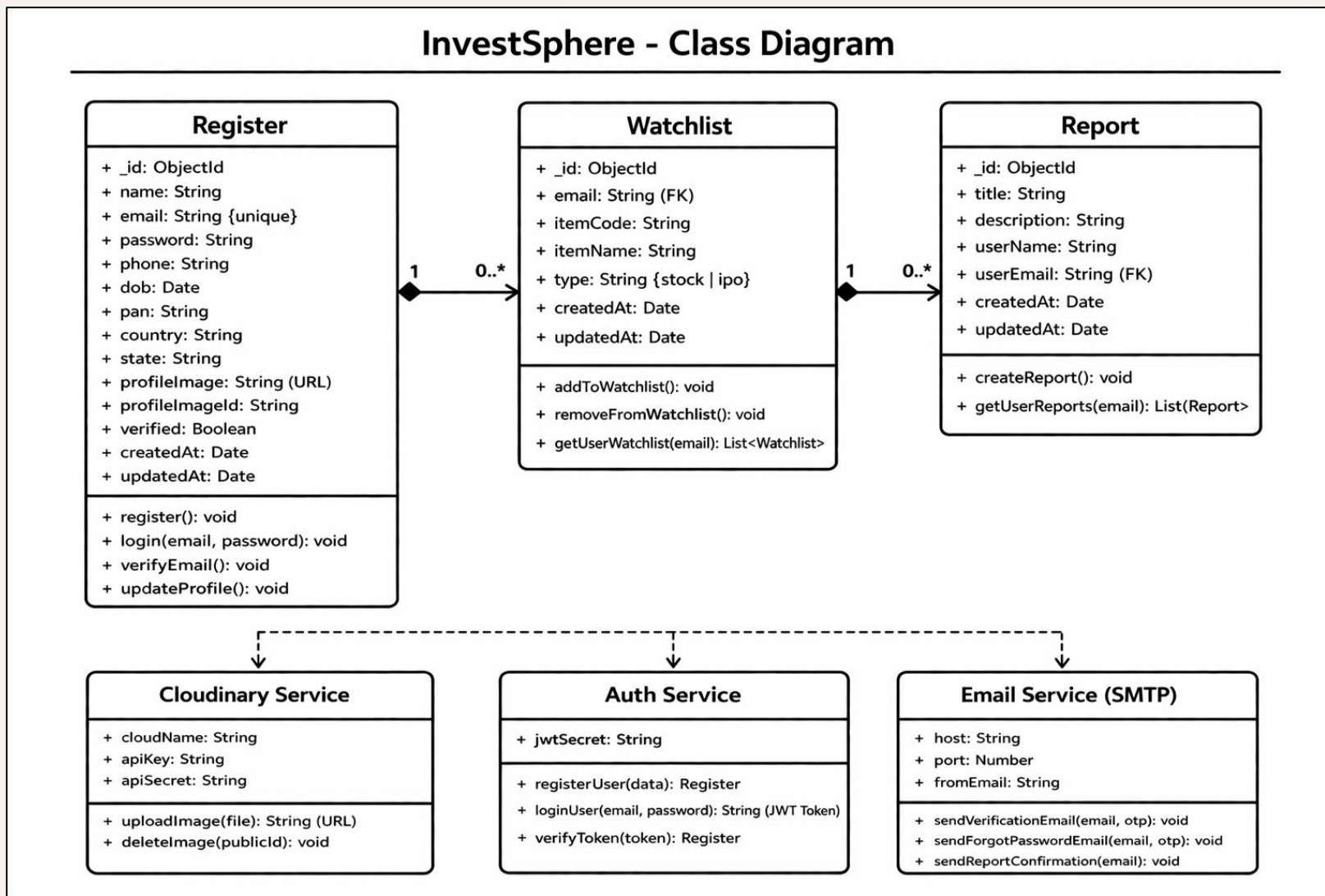


Cloudinary
Media Storage

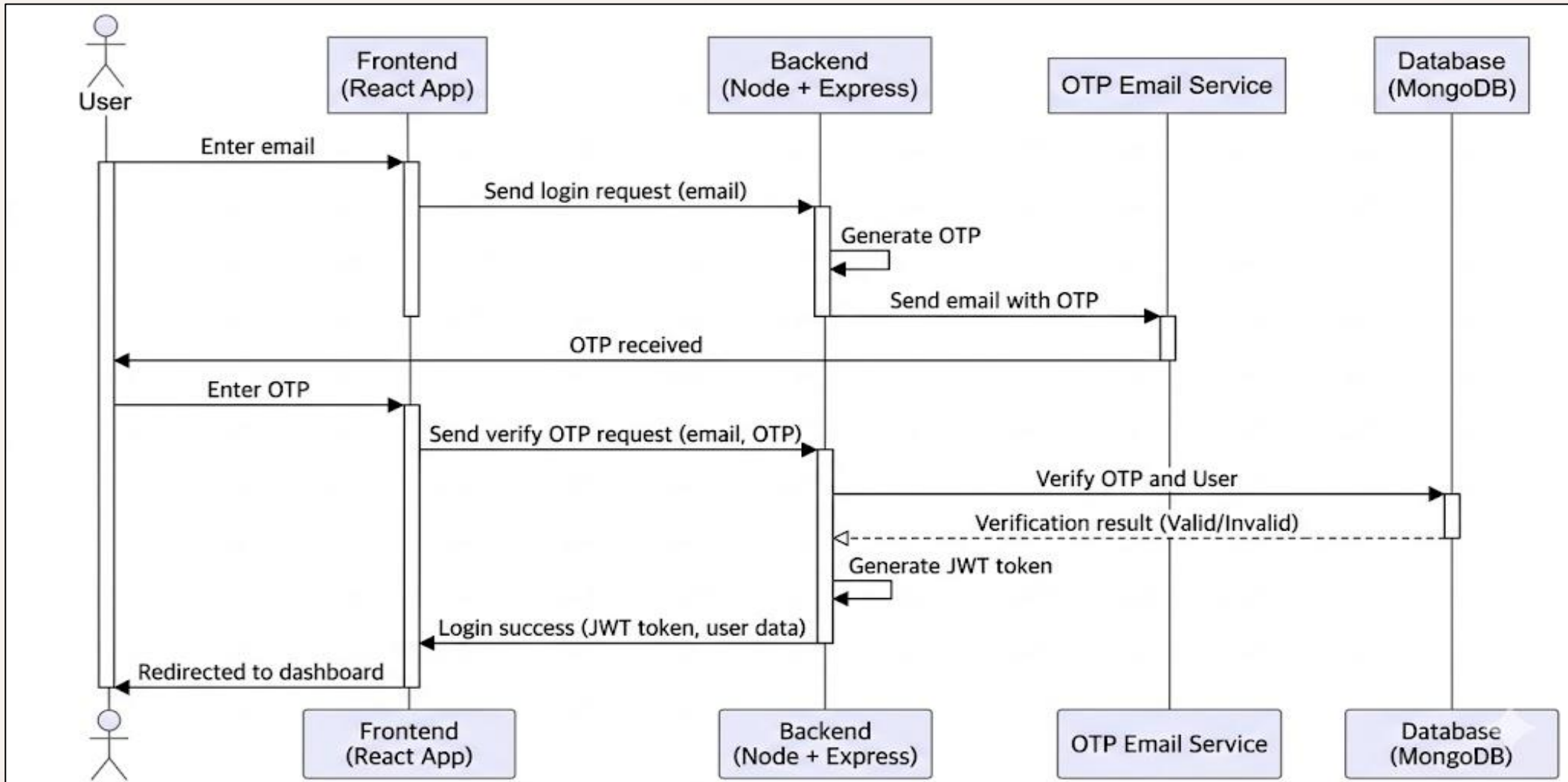
Use Case Diagram:



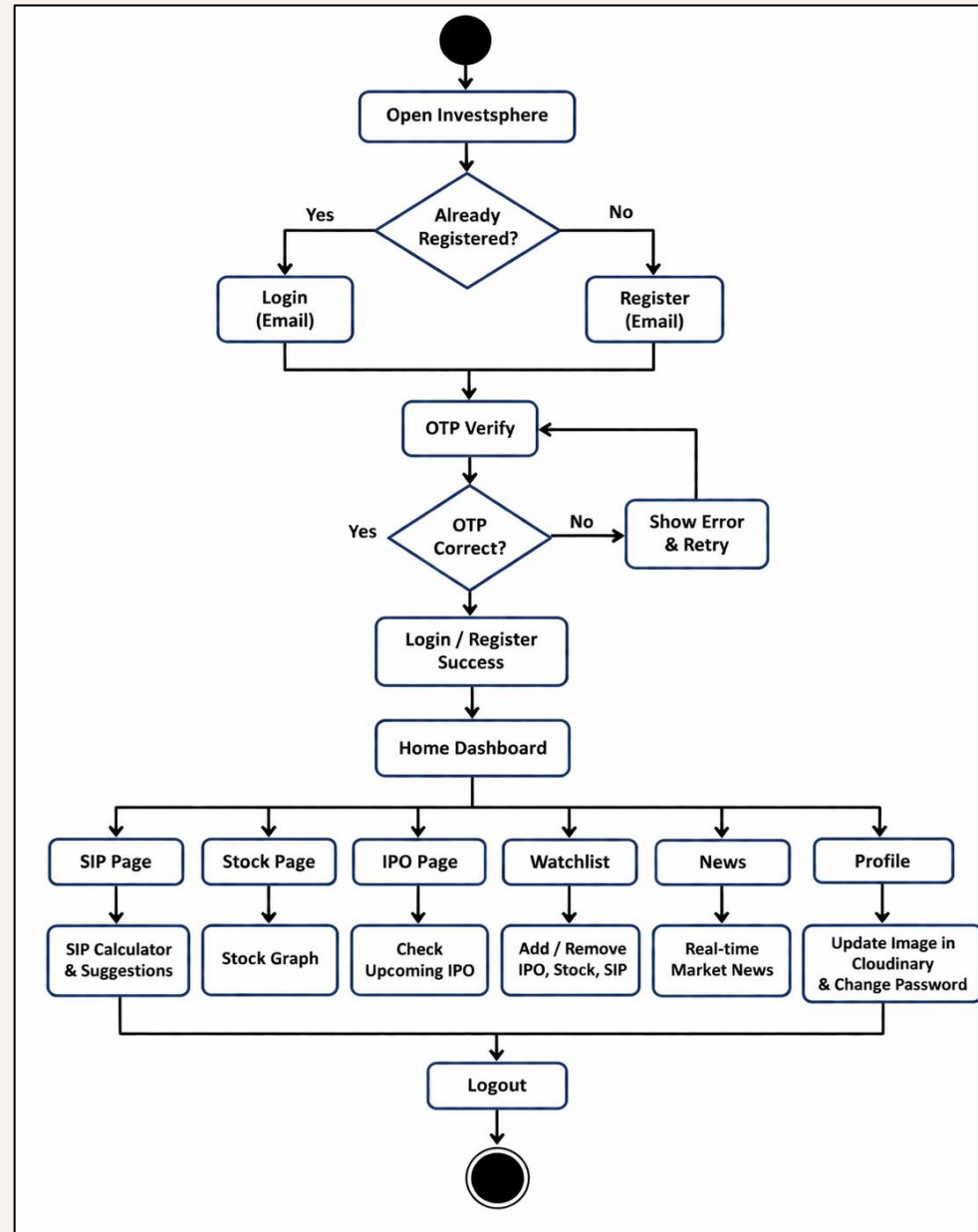
Class Diagram:



Interaction Diagram:



Activity Diagram:



Data Dictionary:

1. User Collection

- Stores user account details like name, email, encrypted password, and profile image (Cloudinary).
- Used for authentication, authorization, and maintaining user identity in the system.

2. Investment Collection

- Tracks all user investments including IPO, Stock, and SIP with amount and timestamps.
- Helps users manage and monitor their portfolio in a centralized dashboard.

3. Watchlist Collection

- Stores assets that users want to track for future investment decisions.
- Allows easy monitoring of IPOs, stocks, and SIPs without investing.

Screenshots:

Registration Page:

Create Account

Register your InvestSphere account

Full Name

Enter your full name

Email Address

Enter email address

Mobile Number

Enter 10 digit mobile nun

Country

India

State

Enter your state

PAN Number

ABCDE1234F

Date of Birth

dd-mm-yyyy

Password

Minimum 6 characters

Confirm Password

Re-enter password

☐ I'm not a robot


reCAPTCHA
[Privacy](#) - [Terms](#)

Register

Already have an account? [Login](#)

Login Page:

InvestSphere
Sign in to your account

Login

[Forgot Password](#)[Forgot Email](#)

Don't have account? [Register](#)

InvestSphere
Sign in to your account

Find Email

Find Emails

Close

[Forgot Password](#)[Forgot Email](#)

Don't have account? [Register](#)

InvestSphere
Sign in to your account

Reset Password

Send OTP

Close

[Forgot Password](#)[Forgot Email](#)

Don't have account? [Register](#)

User Profile Page:

**InvestSphere**

Home

SIP

Stocks

IPO

Watchlist

Market News

Welcome 
Anant Soni



Anant Soni

mranantsoni7@gmail.com

Name

Anant Soni

Phone

7878787878

Country

India

DOB

10/7/2004

Email

mranantsoni7@gmail.com

PAN

GFNPM4206F

State

Gujarat

Verification

Verified

Account Created: 14/3/2026

Account Updated: 27/3/2026

Home Page:

**InvestSphere**

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MSFT
\$359.13
+0.66%

GOOGL
\$273.92
-0.15%

TSLA
\$356.86
-1.37%

AMZN
\$201.87
+1.27%

META
\$537.39
+2.22%

NVDA
\$166.02
-0.90%

NFLX
\$93.18
-0.27%

India's Smart Investment Platform

Smart Investing Starts Here

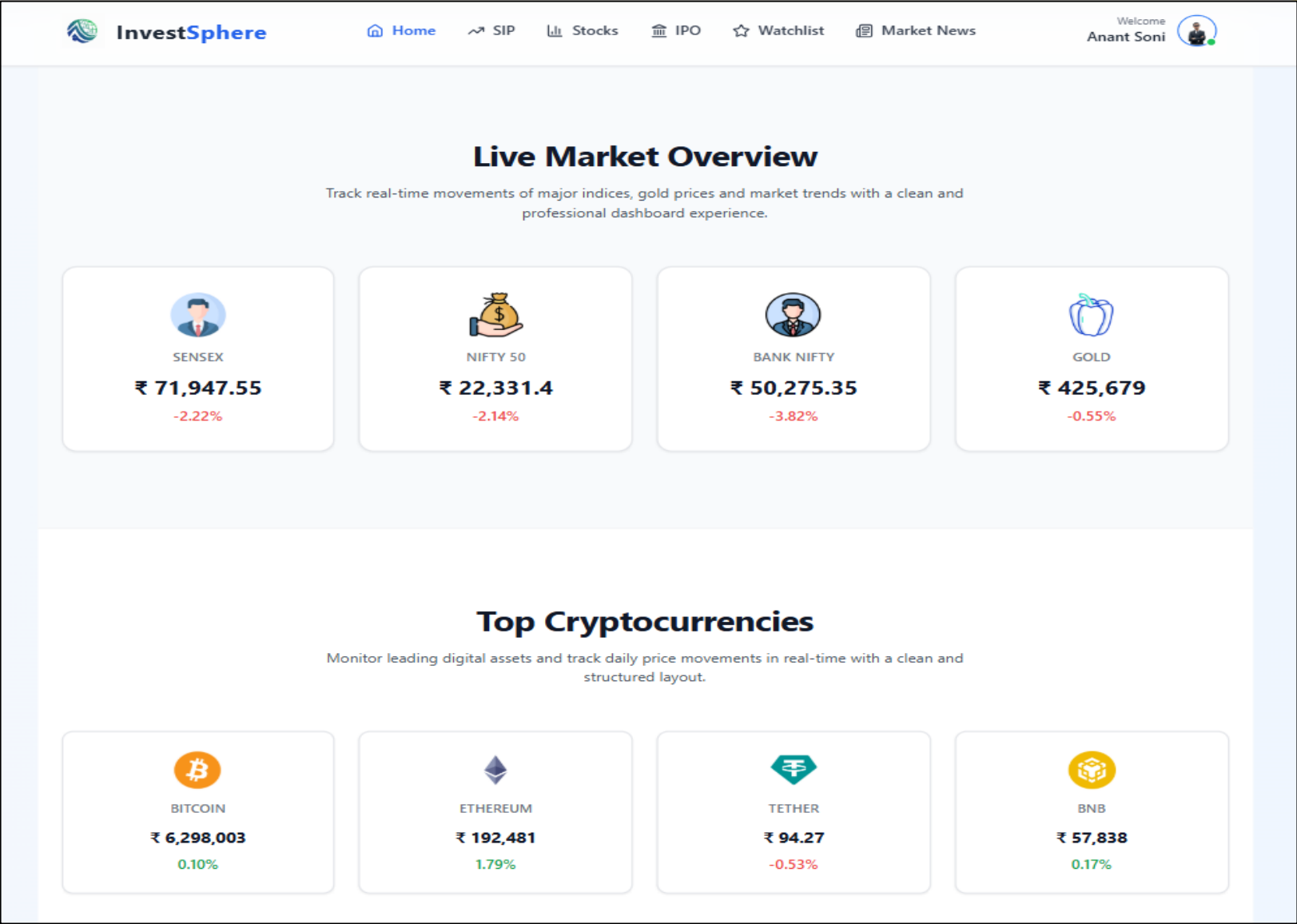
Track markets, explore IPOs, analyze stocks, and plan SIP investments — all from one intelligent financial dashboard built for modern investors.

Explore IPOs

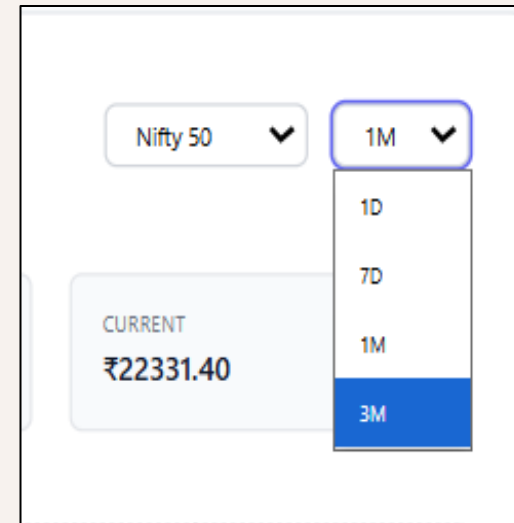
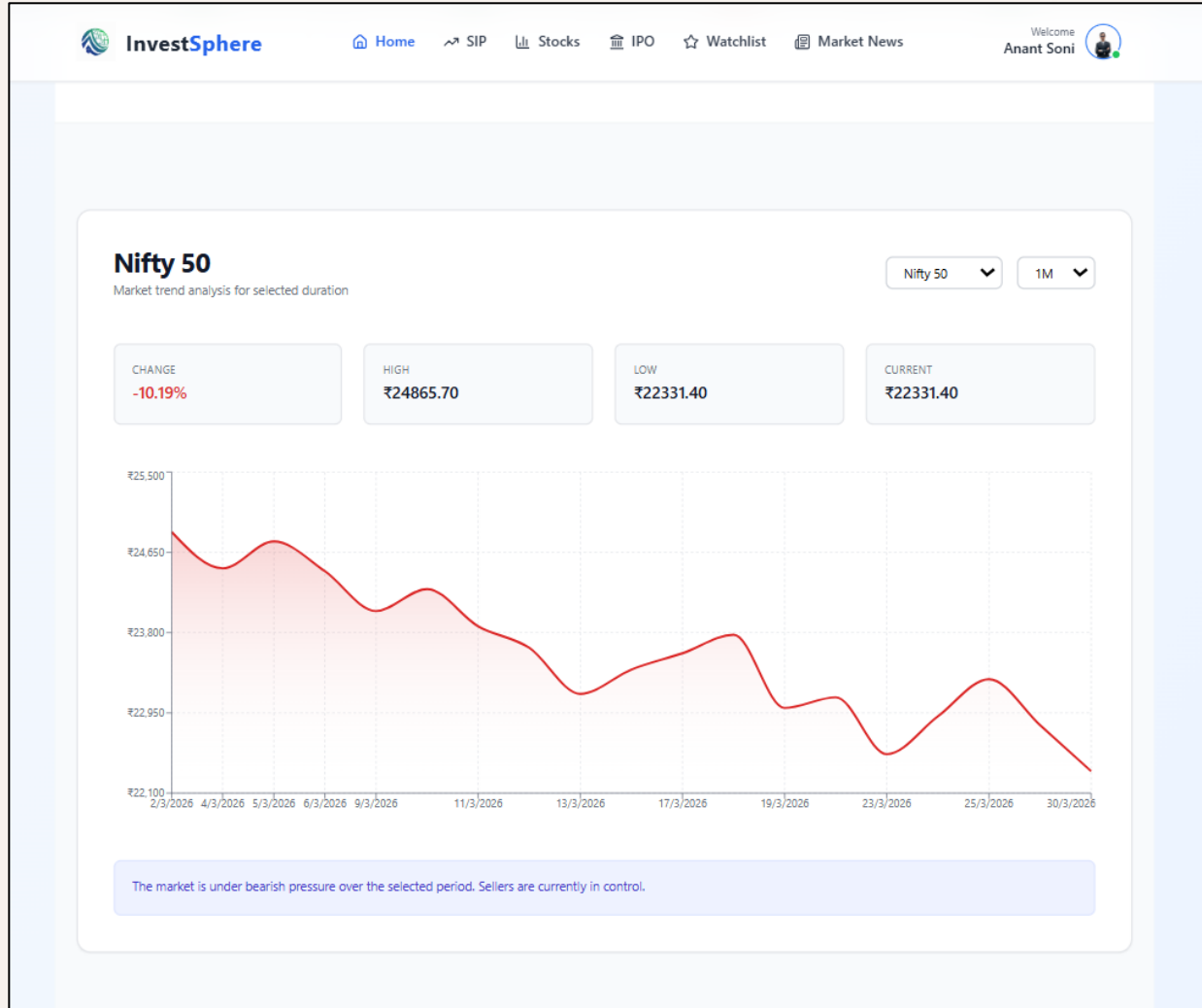
Analyze Stocks

SIP Calculator

Home Page:



Home Page:



SIP Page:



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SIP Investment Funds

Open Smart SIP Planner

SIP Investment Funds

Invest systematically and build long-term wealth with top-rated funds.

Total SIP Funds
11299

Starting SIP
500

Categories
Equity / Debt / Hybrid

Search SIP Fund

Search SIP Fund...

Category

All Equity Debt Hybrid

Risk Level

All Low Medium High

Grindlays Super Saver Income Fund-GSSIF-Half Yearly...

Scheme Code: 100027

Category: Debt

Min SIP: ₹500

Risk: Low

Already in Watchlist View Details

Grindlays Super Saver Income Fund-GSSIF - ST-Dividend

Scheme Code: 100031

Category: Debt

Min SIP: ₹500

Risk: Low

Already in Watchlist View Details

Birla Sun Life Freedom Fund-Plan A (Dividend)

Scheme Code: 100035

Category: Equity

Min SIP: ₹500

Risk: High

Already in Watchlist View Details



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Category: Debt

Min SIP: ₹500

Risk: Low

Already in Watchlist View Details

Aditya Birla Sun Life Income Fund - Regular - Quarterly...

Scheme Code: 100037

Category: Debt

Min SIP: ₹500

Risk: Low

Add to Watchlist View Details

Aditya Birla Sun Life Liquid Fund -weekly IDCW

Scheme Code: 100051

Category: Debt

Min SIP: ₹500

Risk: Low

Already in Watchlist View Details

Category: Debt

Min SIP: ₹500

Risk: Low

Already in Watchlist View Details

Aditya Birla Sun Life Liquid Fund -Institutional - IDCW

Scheme Code: 100041

Category: Debt

Min SIP: ₹500

Risk: Low

Add to Watchlist View Details

Aditya Birla Sun Life Gilt Plus - Liquid Plan - Growth - Regular...

Scheme Code: 100055

Category: Debt

Min SIP: ₹500

Risk: Low

Already in Watchlist View Details

Category: Equity

Min SIP: ₹500

Risk: High

Already in Watchlist View Details

Aditya Birla Sun Life Liquid Fund - Growth

Scheme Code: 100047

Category: Debt

Min SIP: ₹500

Risk: Low

Add to Watchlist View Details

Aditya Birla Sun Life Govenment Securities Fund -Regular - ...

Scheme Code: 100057

Category: Equity

Min SIP: ₹500

Risk: High

Already in Watchlist View Details

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Stock Page:



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Welcome Anant Soni

Explore Stock Market

Discover top performing stocks, track live prices, analyze market trends and explore investment opportunities across Indian markets.

1800+

Listed Stocks

NSE

Exchange

Real Time

Market Data

Search stock...

AllTop GainersTop LosersMost Active52W High52W Low

Top Gainers

SPAISA-RE

+50.00%

3PLAND

+2.72%

3IINFOLD

+1.93%

AAATECH

+1.03%

AADHARHFC

+1.64%

Top Losers

AARTECH

-11.90%

AAREYDRUGS

-9.21%

AARON

-8.71%

20MICRONS

-5.39%

AZZINFRA

-4.97%



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Top Gainers

SPAISA-RE

+50.00%

3PLAND

+2.72%

3IINFOLD

+1.93%

AAATECH

+1.03%

AADHARHFC

+1.64%

Top Losers

AARTECH

-11.90%

AAREYDRUGS

-9.21%

AARON

-8.71%

20MICRONS

-5.39%

AZZINFRA

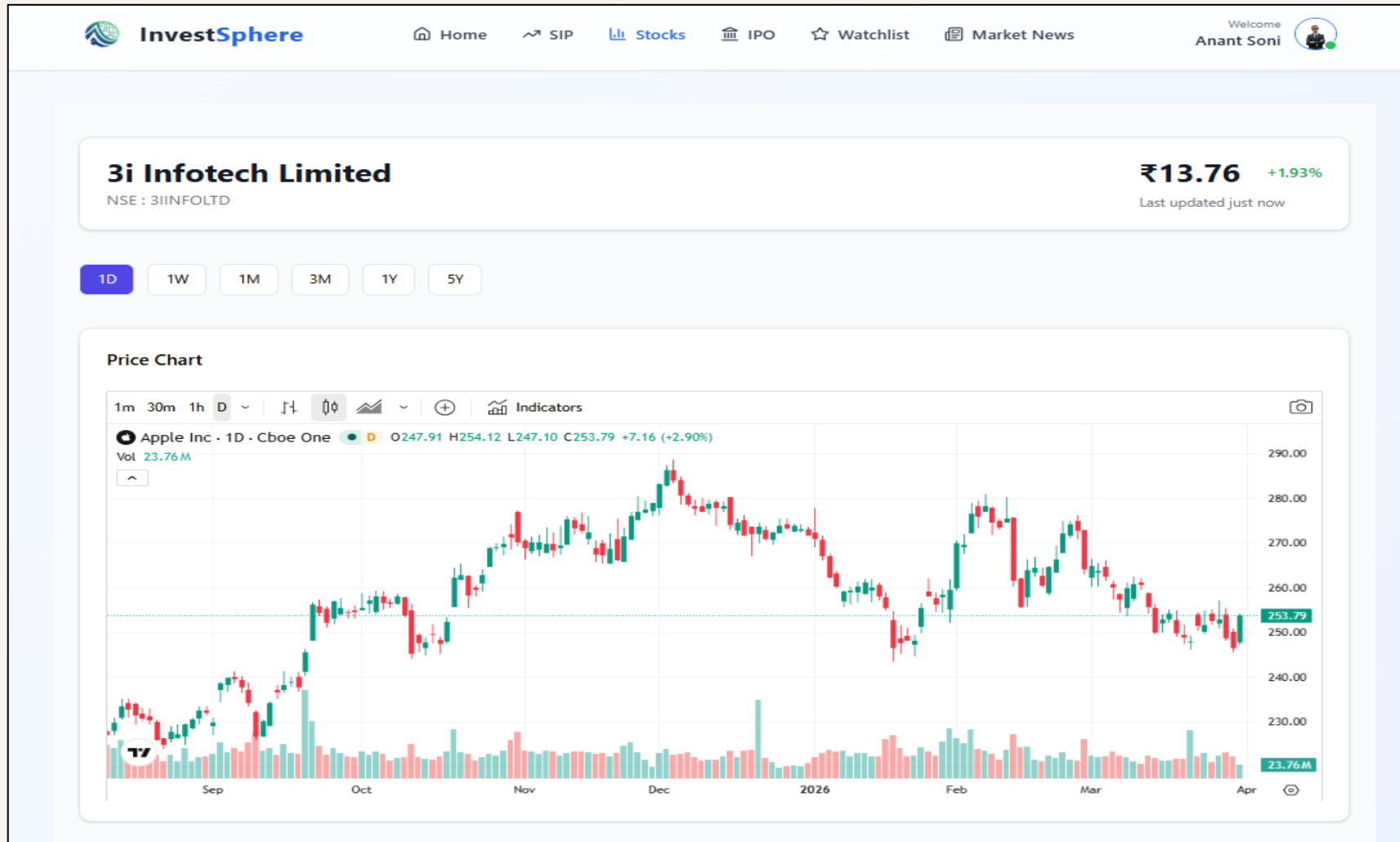
-4.97%

Company	Symbol	Price	Change	Watchlist
Aaron Industries Limited NSE	AARON	₹109.27	-8.71%	Already Added
Aartech Solonics Limited NSE	AARTECH	₹34.80	-11.90%	Add to Watchlist
Aarti Drugs Limited NSE	AARTIDRUGS	₹319.75	-3.31%	Already Added
Aarti Industries Limited NSE	AARTIIND	₹399.05	-4.40%	Add to Watchlist
Aarti Pharmalabs Limited NSE	AARTIPHARM	₹591.95	-2.78%	Add to Watchlist

1

2

Stock Page:



IPO Page:



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[← Back to IPO](#)



IPO Application Process

Follow this step-by-step real-life process to successfully apply for an IPO through your broker app using UPI mandate.



1 Open a Demat & Trading Account

Open a Demat account with a SEBI-registered broker like Zerodha, Upstox, Angel One etc. Complete KYC verification before applying.



2 Go to IPO Section in Broker App

Login to your broker app → Navigate to IPO section → Select the IPO you want to apply for.



3 Select Investor Category & Lot Size

Choose Retail (₹2 lakh limit) or HNI category. Enter number of lots you want to apply for.



4 Enter UPI ID & Submit Application

Enter your UPI ID correctly. Double-check details before submitting.



5 Approve UPI Mandate

You will receive a mandate request in your UPI app. Approve it within the deadline to block funds.



6 Check Allotment Status

After IPO closes, check allotment on registrar website. If allotted → shares credited. If not → money unblocked.



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5 Approve UPI Mandate

You will receive a mandate request in your UPI app. Approve it within the deadline to block funds.



6 Check Allotment Status

After IPO closes, check allotment on registrar website. If allotted → shares credited. If not → money unblocked.



Important Notes

- UPI mandate approval is mandatory.
- Funds are only blocked, not deducted initially.
- If IPO is oversubscribed, allotment is lottery based.
- Refund happens automatically if not allotted.



Before Applying Checklist

- Active Demat account
- Correct UPI ID linked to bank account
- Enough balance for lot amount
- Verified KYC details

IPO Page:



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IPO Application Process Instructions

Learn step-by-step how to apply for an IPO through your broker app.

Upcoming IPOs

Ongoing IPOs

Closed IPOs

Q Search IPO by company name or symbol...

AllNASDAQNYSE

HMH Holding Inc

NASDAQ Global Select

Symbol: HMH

Exchange: NASDAQ Global Select

Open Date: 2026-04-01

Close Date: 2026-04-03

Issue Price: 19.00-22.00

Shares Offered: 1,05,20,000

Total Value: ₹ 26,61,56,000

Upcoming

Already in Watchlist



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Learn step-by-step how to apply for an IPO through your broker app.

Upcoming IPOs

Ongoing IPOs

Closed IPOs

Q Search IPO by company name or symbol...

AllNASDAQNYSE

Live Market IPO Ltd.

NSE

Symbol: DEMO

Exchange: NSE

Open Date: 2026-03-28

Close Date: 2026-03-30

Issue Price: ₹ 125 - 135

Shares Offered: 25,00,000

Total Value: ₹ 33,75,00,000

Ongoing

Apply Now

Add to Watchlist

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IPO Application Process Instructions

Learn step-by-step how to apply for an IPO through your broker app.

[Upcoming IPOs](#) [Ongoing IPOs](#) [Closed IPOs](#)

[All](#) [NASDAQ](#) [NYSE](#)

Apogee Acquisition Corp
NASDAQ Global

Symbol: AACPU
Exchange: NASDAQ Global
Open Date: 2026-03-23
Close Date: 2026-03-27
Issue Price: 10.00
Shares Offered: 2,50,00,000
Total Value: ₹ 28,75,00,000

Closed

Inflection Point Acquisition Corp. VI
NASDAQ Global

Symbol: IPFXU
Exchange: NASDAQ Global
Open Date: 2026-03-23
Close Date: 2026-03-27
Issue Price: 10.00
Shares Offered: 2,20,00,000
Total Value: ₹ 25,30,00,000

Closed

Future Money Acquisition Corp
NASDAQ Global

Symbol: FMACU
Exchange: NASDAQ Global
Open Date: 2026-03-23
Close Date: 2026-03-27
Issue Price: 10.00
Shares Offered: 1,00,00,000
Total Value: ₹ 11,50,00,000

Closed

QDRO Acquisition Corp.
NASDAQ Global

Symbol: QADRU
Exchange: NASDAQ Global
Open Date: 2026-03-23
Close Date: 2026-03-27
Issue Price: 10.00
Shares Offered: 2,00,00,000
Total Value: ₹ 23,00,00,000

Closed

Maywood Acquisition Corp. 2
NASDAQ Global

Symbol: MYXXU
Exchange: NASDAQ Global
Open Date: 2026-03-22
Close Date: 2026-03-26
Issue Price: 10.00
Shares Offered: 1,00,00,000
Total Value: ₹ 11,50,00,000

Closed

Janus Living, Inc.
NYSE

Symbol: JAN
Exchange: NYSE
Open Date: 2026-03-16
Close Date: 2026-03-20
Issue Price: 20.00
Shares Offered: 4,20,00,000
Total Value: ₹ 84,00,00,000

Closed

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How IPO Works?

Understand the basics of IPO investing before applying. A clear understanding helps you make smarter investment decisions.

What is an IPO?
IPO (Initial Public Offering) is when a private company offers its shares to the public for the first time to raise capital.

Retail vs HNI vs QIB
Retail: Up to ₹2 Lakhs
HNI: Above ₹2 Lakhs
QIB: Mutual Funds & Banks

What is GMP?
Grey Market Premium indicates unofficial demand before listing. It gives an idea of expected listing price.

Listing Gain
Listing gain is profit earned when stock lists above its IPO issue price.

Allotment Process
Shares are allotted based on subscription levels. Oversubscription leads to lottery system for retail investors.

Why Companies Launch IPO?
Companies launch IPOs to raise capital for expansion, repay debts, improve brand credibility, and allow early investors to exit.

IPO investments are subject to market risks. Always read the prospectus carefully before investing.

Watchlist Page: (Stock Watchlist)

 InvestSphere

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 **Your Watchlist**

Track your favourite Stocks, SIP funds and IPOs.

[Stock Watchlist](#)

[SIP Watchlist](#)

[IPO Watchlist](#)

Company	Symbol	Price	Change	Action
Aarti Drugs Limited NSE	AARTIDRUGS	₹319.75	-3.31%	 Remove
3P Land Holdings Limited NSE	3PLAND	₹29.50	+2.72%	 Remove
360 ONE WAM LIMITED NSE	360ONE	₹949.60	-2.84%	 Remove
Aaron Industries Limited NSE	AARON	₹109.27	-8.71%	 Remove
Aarnav Fashions Limited NSE	AARNAV	₹21.69	-4.91%	 Remove

Watchlist Page: (SIP Watchlist)

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Welcome **Anant Soni**

☆ **Your Watchlist**

Track your favourite Stocks, SIP funds and IPOs.

[Stock Watchlist](#)

[SIP Watchlist](#)

[IPO Watchlist](#)

HDFC Balanced Advantage Fund - Growth Plan

SIP

Scheme Code: 100119

Category

Hybrid

Min SIP

₹5,000

Risk

Medium

Remove

View Details

DSP Aggressive Hybrid Fund - Regular Plan - Growth

SIP

Scheme Code: 100081

Category

Hybrid

Min SIP

₹500

Risk

Medium

Remove

View Details

Aditya Birla Sun Life Gilt Plus - Liquid Plan - Growth -...

SIP

Scheme Code: 100055

Category

Debt

Min SIP

₹500

Risk

Low

Remove

View Details

Aditya Birla Sun Life Liquid Fund -weekly IDCW

SIP

Scheme Code: 100051

Category

Debt

Min SIP

₹500

Risk

Low

Remove

View Details

Aditya Birla Sun Life Government Securities Fund ...

SIP

Scheme Code: 100057

Category

Equity

Min SIP

₹500

Risk

High

Remove

View Details

Grindlays Super Saver Income Fund-GSSIF-Half Yearly...

SIP

Scheme Code: 100027

Category

Debt

Min SIP

₹500

Risk

Low

Remove

View Details

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Watchlist Page: (IPO Watchlist)

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Welcome 
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 **Your Watchlist**

Track your favourite Stocks, SIP funds and IPOs.

 Stock Watchlist

 SIP Watchlist

 **IPO Watchlist**

HMH Holding Inc

NASDAQ Global Select

Symbol: HMH

Open Date: 2026-04-01

Close Date: 2026-04-03

Issue Price: 19.00-22.00

upcoming

Apply Now

Remove

Metals Royalty Co Inc.

NSE

Symbol: TMCR

Open Date: TBA

Close Date: TBA

Issue Price: TBA

upcoming

Apply Now

Remove

Market News Page:



HomeSIPSIPStocksIPOWatchlistMarket News

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Market Insights & Financial News

Stay updated with latest IPO, SIP, Stock & Market trends.

Market

IPO

SIP

Stock



Top stocks to buy: Stock recommendations for March 30, 2026 week

3/30/2026

India Business News: Stock market recommendations: Ipca Laboratories, and AU Small Finance Bank are the stocks that Motilal Oswal...

Read Full Article



Stock Market Crash: Rs 13 Lakh Crore Gone; What Triggered Today's Wealth Wipeout...

3/23/2026

The fall was widespread, with all 30 Sensex stocks trading in the red. Heavyweights like Tata Steel, State Bank of India, HDFC Bank, Bajaj Finance,...

Read Full Article



Stock Market Today Highlights: Sensex up 380 points, Nifty above 23,100 at close

3/20/2026

Stock Market Today Highlights: At close, Nifty was above 23,100 while Sensex up by 380 points. In terms of sectors, all other sectoral indices ended i...

Read Full Article



Coal India subsidiary CMPDI share price crashes 9% after weak listing. Buy, sell or...

3/30/2026

CMPDI IPO listing took place amid the stock market crash today, as the benchmark indices, Sensex and Nifty 50 declined over 1.5% each....

Read Full Article



India's biggest IPO now trades 10% below its issue price, posts worst month since...

3/30/2026

Shares of Hyundai Motor India have declined over 19% so far in the month of March, which is the worst month that the stock has had, since its...

Read Full Article



Central Mine IPO listing in focus. What does GMP signal ahead of debut?

3/27/2026

Central Mine IPO is all set to make its stock market debut on Monday. The grey market premium signals a positive listing for the Coal India...

Read Full Article













API References & Integrations

1. Crypto & Gold API <https://api.coingecko.com/api/v3/coins/markets>
2. IPO Data API - <https://finnhub.io/api/v1/calendar/ipo>
3. Stock Market Data API - <https://query1.finance.yahoo.com/v8/finance/chart>
4. SIP / Mutual Fund API - <https://api.mfapi.in/mf>
5. Market News API - <https://gnews.io/api/v4/search>
6. NSE Stock List Data - https://archives.nseindia.com/content/equities/EQUITY_L.csv
7. Cloud Storage (Images) - <https://console.cloudinary.com/>

Additional Reference Platforms

1. Grow Platform - <https://groww.in/>
2. Angel One - <https://www.angelone.in/>
3. Moneycontrol - <https://www.moneycontrol.com/>
4. Yahoo Finance - <https://finance.yahoo.com/>
5. Upstox - <https://upstox.com/>

The background is white with decorative blue and yellow geometric shapes in the corners. The top-left corner has a blue shape, the top-right has a yellow shape, the bottom-left has a yellow shape, and the bottom-right has a blue shape.

Thank You.